How Much Interval Width Is Left in a Fixed Prediction Score? A Paired Study of Nonlinear Controls and Coverage Costs

Autonomous research report
Reproducible CPU experiments and independent analysis

Abstract

Can inference improve when labels, unlabeled observations, and a pretrained predictor cannot change? We study population means using only the predictor's frozen scalar score. Elementary control-variate analysis identifies the unused information: the part of the conditional label mean not explained by an affine function of the score. We compare scalar PPI++, cubic score controls, and cross-fitted controls on eight synthetic laws and three public prediction artifacts, using paired repeated sampling and Monte Carlo uncertainty. At 800 labels, cross-fitted cubic controls reduce average Wald width by 69.0% in a low-noise quadratic example and 15.4% in a binary quadratic example; their coverage is consistent with 95% at the reported Monte Carlo resolution. Extra gains on the real artifacts are negligible. Unprotected high-dimensional controls and rare-event Wald intervals under-cover. For a literal finite-sample guarantee, a pilot-fitted, bounded control combined with empirical-Bernstein bounds reduces width by 28.3% against the label-only empirical-Bernstein interval in the continuous quadratic example at 12,800 labels, but is wider at 800. Thus fixed inputs leave conditional opportunities, not a general promise of shorter valid intervals. The contribution is a quantitative, reproducible audit of this tradeoff, not a new general theory of calibration or semisupervised efficiency.

1 Introduction

Costly labels and abundant model predictions motivate prediction-powered inference (PPI): use predictions to reduce variance and a labeled correction to protect the scientific target [Angelopoulos et al., 2023]. The practical question here is retrospective. An analyst already has a labeled set, an unlabeled set, and a predictor that cannot be retrained. Could changing only the inference procedure yield a shorter valid confidence interval?

We choose the mean $\mu = \mathbb{E}Y$. It represents prevalence for binary labels and a scientific average for bounded measurements; it also permits transparent, exact variance comparisons. We restrict attention to the scalar score $S = f(X)$, even when extra features are available. The original function f and every one of its scores remain unchanged. Prespecified functions of S and coefficients estimated from the existing labels are components of the inference procedure, just as the scalar tuning coefficient in PPI++ is. If even such inference coefficients were forbidden, the admissible procedure class would be narrower than the one studied here.

"Fixed" means that every competing procedure receives the *same realized inputs within a repetition*; it does not mean conditioning a frequentist coverage claim on all those inputs. Coverage is evaluated over repeated sampling of labeled pairs and unlabeled scores. Without an additional conditional

model, realized-data conditional coverage or realized-data minimum width does not follow from a marginal confidence statement.

Our questions are: (i) what information can scalar PPI++ leave unused in S; (ii) can a low-dimensional, label-fitted adjustment exploit it without additional data; and (iii) how much of the width improvement survives an unconditional finite-sample coverage requirement? Our contributions are a projection-based accounting of the possible gain and its estimation cost; a paired benchmark with positive, null, and failure cases; and a direct finite-sample comparison including exact binomial baselines. We explicitly separate familiar methodology from the empirical contribution. We do not establish a procedure that dominates all existing valid intervals, and we do not claim a comprehensive literature review beyond the permitted sources.

2 Related work and scope

PPI estimates a mean with $\bar{S}_U + \overline{Y - S}_L$ and provides a broader rectifier framework [Angelopoulos et al., 2023]. PPI++ learns a scalar coefficient, reducing dependence on the raw predictor's calibration [Angelopoulos et al., 2024]. Its discussion already connects prediction-powered means to augmented inverse-propensity estimation and gives an efficiency inequality. Our projection calculations specialize this existing perspective to the information in a frozen score and expose exactly what nonlinear controls could add. A multivariate control variate is not a newly invented estimator.

Cross-PPI uses the available labels to train predictors with cross-fitting [Zrnic and Candès, 2024b]. Here we train no replacement model on X: the nuisance regressions operate only on a fixed, low-dimensional score dictionary. The motivation for cross-fitting is shared with double/debiased machine learning [Chernozhukov et al., 2024], but our asymptotic argument requires only a fixed-dimensional consistent coefficient, not a general causal or high-dimensional nuisance theorem. StratPPI accommodates strata and allocation choices [Fisch et al., 2024]. Retrospectively choosing different labels is excluded in our experiment; a fixed-bin poststratification estimator is closely related to a score dictionary. Active inference changes which points receive labels [Zrnic and Candès, 2024a] and is therefore outside our constraint. We make no comparison to its altered sampling designs. Our strict-coverage construction applies the empirical-Bernstein inequality of Maurer and Pontil [2009]; its conservatism is measured rather than hidden.

3 What information can still reduce width?

Assume $(Y_i, S_i)_{i=1}^n$ are IID from P, $(\widetilde{S}_j)_{j=1}^N$ are independently IID from P_S, and f is fixed independently of these observations. The target is the superpopulation mean, not the mean of the observed unlabeled pool. Define $\rho = N/(n+N)$ and a fixed dictionary $\phi(S) \in \mathbb{R}^d$ with covariance Σ and $c = \mathrm{Cov}(\phi(S), Y)$. For deterministic b, use

$$\widehat{\mu}_b = \bar{Y}_L + b^\top(\bar{\phi}_U - \bar{\phi}_L), \qquad V(b) = \frac{\mathrm{Var}(Y - b^\top\phi)}{n} + \frac{\mathrm{Var}(b^\top\phi)}{N}. \tag{1}$$

It is unbiased for every b. The distinction between unbiasedness for a fixed b and a data-dependent fitted $\widehat{b}$ matters at small n.

Proposition 1 (Projection headroom)**.** *Suppose second moments exist and Σ is nonsingular. Write $\Pi_d m$ for the centered $L^2(P_S)$ projection of $m(S) = \mathbb{E}[Y \mid S]$ onto the centered dictionary. Then*

$$b_d^* = \rho\Sigma^{-1}c, \qquad V_d^* = \frac{1}{n}\{\mathrm{Var}\, Y - \rho c^\top\Sigma^{-1}c\}, \tag{2}$$

$$V(b) - V_d^* = (n^{-1} + N^{-1})(b - b_d^*)^\top\Sigma(b - b_d^*). \tag{3}$$

For nested dictionaries, the variance gain is ρ/n times the added squared projection norm. Scalar PPI++ uses only the affine score span. If $\sigma_m^2 = \mathrm{Var}\, m(S)$ and $\sigma_\epsilon^2 = \mathbb{E}\,\mathrm{Var}(Y \mid S)$, the unrestricted square-integrable score-control optimum is

$$V_S^* = \frac{\sigma_\epsilon^2}{n} + \frac{\sigma_m^2}{n+N}, \qquad V_1^* - V_S^* = \frac{\rho}{n}\mathbb{E}[(m - \Pi_{\mathrm{aff}} m)^2]. \tag{4}$$

Here Π_{aff} includes the intercept. The oracle Wald width ratio is $\sqrt{V_S^*/V_1^*}$.

A proof by completing the square appears in Appendix A. Equation (4) states both an opportunity and a limit. If m is affine, there is no first-order score-only gain beyond optimally tuned scalar PPI++. A two-valued score always has an affine conditional mean on its support. If Y is mean-independent of S, scores cannot reduce first-order variance. Extra transformations cannot create label information, and estimated extra coefficients can cost accuracy.

Proposition 2 (Regular score-only lower bound)**.** *In the unrestricted IID model for (Y, S), with finite second moments and n/N tending to a positive finite constant, V_S^* is the first-order variance lower bound among pathwise regular asymptotically linear mean estimators using only these labeled pairs and scores.*

This is a regular-estimator bound, not a finite-sample minimum-length theorem and not a ban on gains from extra X features, structural assumptions, or nonregular procedures. In particular, it does not prove that a chosen realized interval cannot be shortened. Appendix A derives the bound directly from the influence-function constraint.

4 Inference procedures and their guarantees

Full-sample and cross-fitted controls. On $S \in [0,1]$, we prespecify $\phi_d(S) = (P_1(2S-1), \ldots, P_d(2S-1))$, where P_k are Legendre polynomials. CV3 uses $\widehat{b} = \rho\widehat{\Sigma}^{+}_{L\cup U}\widehat{c}_L$ with $d = 3$; the superscript denotes a Moore–Penrose inverse. Scalar PPI++ is the same construction with $d = 1$. We show its unrestricted version as the main scalar baseline and separately report clipping the coefficient of the raw score to $[0,1]$, as in the reference implementation. All methods use unbiased sample variances; this documented divisor choice differs slightly from the package's mean-interval code.

CF3 instead fits an intercept and cubic dictionary by ordinary least squares on four of five fixed, equal folds of the labeled data. In held-out fold k, let $\widehat{b}_{-k}$ be ρ times the fitted slope. The estimator is

$$\widehat{\mu}_{\mathrm{CF}} = \bar{Y}_L + \frac{1}{K}\sum_{k=1}^{K} \widehat{b}_{-k}^{\top}(\bar{\phi}_U - \bar{\phi}_{L_k}), \quad K = 5. \tag{5}$$

CF1 uses only the affine score and isolates the benefit of nonlinear information from differences in coefficient estimation. Each method receives the same labels and scores; folds use no extra labels. There is no model selection on the reported test repetitions.

Proposition 3 (Fixed-dimensional Wald validity)**.** *Assume a fixed dictionary of finite fourth moments, a positive definite covariance on its nonconstant span, positive limiting variance, and $n/N \to r \in (0,\infty)$. Then CV3 and the fixed-fold OLS control estimator are asymptotically normal with variance V_3^*. With the variance estimates below, their nominal Wald intervals have limiting coverage $1-\alpha$.*

For CV3 use $\widehat{V} = s_L^2(Y - \widehat{b}^{\top}\phi)/n + s_U^2(\widehat{b}^{\top}\phi)/N$. For CF3 use

$$\widehat{V}_{\mathrm{CF}} = \sum_k \frac{|L_k|}{n^2} s^2_{L_k}(Y - \widehat{b}_{-k}^{\top}\phi) + \frac{s_U^2(\bar{b}^{\top}\phi)}{N}, \qquad \bar{b} = K^{-1}\sum_k \widehat{b}_{-k}. \tag{6}$$

The same unlabeled sample is shared by folds, so its variance uses the *average* coefficient, rather than averaging independent-fold variances. We report $\widehat{\mu} \pm z_{.975}\sqrt{\widehat{V}}$ without clipping for Wald methods. None of these normal approximations has a uniform finite-sample guarantee, particularly for rare labels or a large dictionary.

A literal finite-sample guarantee. For bounded IID variables of known range length R, define

$$B_q(v, R, \delta) = \sqrt{\frac{2v\log(4/\delta)}{q}} + \frac{7R\log(4/\delta)}{3(q-1)}, \quad q \geq 2, \tag{7}$$

where v is the unbiased sample variance. A two-sided application of Maurer and Pontil [2009, Theorem 4] bounds the absolute mean error by B_q with failure probability at most δ.

Split off the first $t = \lfloor n/5 \rfloor$ labels as a pilot. Fit the same degree-d OLS regression on this pilot only, leaving $\ell = n - t$ labels for inference. Set $\rho_\ell = N/(N+\ell)$ and $g(s) = \rho_\ell \,\mathrm{clip}_{[0,1]}(\widehat{m}_d(s))$. EB-split$d$ centers an interval at $\overline{Y-g}_I + \bar{g}_U$ with half-width

$$B_\ell(s_I^2(Y-g), 1+\rho_\ell, \alpha/2) + B_N(s_U^2(g), \rho_\ell, \alpha/2). \tag{8}$$

The original predictor remains fixed; the pilot estimates only nuisance control coefficients. Known bounds on Y and g, rather than observed sample ranges, determine the constants.

Theorem 1 (Finite-sample coverage and split cost)**.** *Assume $Y, S \in [0,1]$, the IID sampling conditions above, and $\ell, N \geq 2$. EB-splitd, after clipping endpoints to $[0,1]$, covers μ with probability at least $1-\alpha$ for every P and any pilot fitting rule. Conditional on the pilot, its point estimator has mean μ and variance*

$$\frac{\sigma_\epsilon^2}{\ell} + \frac{\sigma_m^2}{\ell+N} + (\ell^{-1} + N^{-1})\,\mathrm{Var}\{g(S) - \rho_\ell m(S)\}. \tag{9}$$

Condition on the pilot, apply the two bounds with error $\alpha/2$ each, and use a union bound; integrating over the pilot preserves coverage. The variance identity follows by the same square completion as Proposition 1. Thus label splitting, nuisance estimation, the two square-root terms, and the range penalties all compete with nonlinear headroom. A strict guarantee does not imply a shorter interval.

We compare to EB-label ($B_n(s_Y^2, 1, \alpha)$), EB-raw (the same rectifier bound with $g = S$ and all labels), a label-only Hoeffding interval, and, for known binary outcomes, the exact equal-tailed binomial interval. Using exact binomial intervals when applicable prevents conservative bounded-outcome baselines from creating misleading gains.

5 Experiments

5.1 Design and uncertainty

The main experiment has $n \in \{200, 800\}$, $N = 10n$, and 5,000 independent repetitions per condition. All methods in one repetition receive identical arrays. We use a fixed dictionary and all prespecified laws, including adverse cases. A separate degree stress test uses $d \in \{1, 2, 3, 7, 15\}$ and $n \in \{50, 200, 800\}$. An unlabeled-size check fixes $n = 800$ and uses $N/n \in \{1/4, 1, 4, 40\}$. Strict-coverage checks add $n = 3,200$ and $12,800$, with 3,000 repetitions per condition. This changes sample sizes *between experimental conditions*, never between methods in a comparison.

Synthetic scores are uniform on $[0,1]$, except a two-valued score with equal probabilities. With $T = 2S - 1$, continuous labels have mean $0.5 + 0.4T$ (linear) or $0.1 + 0.8T^2$ (quadratic), plus independent uniform noise on $[-0.03, 0.03]$. Binary laws use success probabilities $0.5 + 0.45T$, $0.05 + 0.9T^2$, $0.05 + 0.9\mathbf{1}\{S > 0.7\}$, 0.5, or 0.005. For two-valued S, the success probability is $0.1 + 0.8S$. All laws are bounded and all targets are evaluated by analytic moment integration in the supplied code. The low-noise quadratic example deliberately isolates strong nonlinear headroom; it is not representative evidence of a typical scientific dataset.

We also use the galaxies, forest, and alphafold prediction/label arrays pinned to the loader in `ppi-python` v0.2.3 [Angelopoulos and contributors, 2024]. Each artifact defines an empirical distribution: independently draw labeled records and unlabeled scores *with replacement*, and evaluate coverage for the full artifact's label mean. This produces IID sampling with known truth and avoids finite-population corrections. It is not coverage evidence for the actual cosmos, Amazon, or proteome. Hidden artifact labels are used only to evaluate truth and oracle projections, never to fit an interval. We ignore the extra protein-modification covariates to maintain the score-only comparison. Scores were already supplied; no predictor was trained or queried.

Coverage uncertainty is binomial Monte Carlo uncertainty, not an additional interval for μ. Main tables show ± 1.96 binomial standard errors; full tables and figures use Wilson intervals. Mean-width error bars are 1.96 standard errors. Width ratios are ratios of mean widths with *paired* delta-method errors: for ratio $R = \bar{W}_A/\bar{W}_B$, use $\mathrm{sd}(W_A - RW_B)/(\sqrt{M}\bar{W}_B)$. These are pointwise summaries, with no multiplicity-adjusted discovery claim. Seeds, exact laws, every per-repetition endpoint, and all results are provided. Implementation checks verify the official scalar tuning formula (accounting for the declared divisor), the projection identities, binary-score equivalence, and input immutability.

5.2 Nonlinear headroom helps only in the right laws

Table 1 identifies conditional, rather than universal, improvements. In the continuous quadratic law, CF3's width ratio to PPI++ is 0.3096 ± 0.0002, with coverage 94.90 [94.25, 95.48]%; the cubic dictionary contains the true mean, while the score's linear covariance with Y vanishes. The binary

Table 1: Synthetic main comparison at $n = 800$, $N = 8,000$, 5,000 paired repetitions. Coverage is percent ± 1.96 Monte Carlo standard errors; width ratios have paired 95% Monte Carlo error bars. CF3 uses cubic controls; CF1 uses only the affine score. Rare-event widths must be interpreted together with their failed coverage.

Law	PPI++ cov. (%)	CF3 cov. (%)	CF3/PPI++	CF3/CF1
Linear, continuous	94.64 ± 0.62	94.62 ± 0.63	0.9967 ± 0.0001	1.0001 ± 5.4e-06
Quadratic, continuous	94.58 ± 0.63	94.90 ± 0.61	0.3096 ± 0.0002	0.3090 ± 0.0002
Linear, binary	95.00 ± 0.60	95.00 ± 0.60	1.0019 ± 0.0001	1.0011 ± 3.7e-05
Quadratic, binary	94.80 ± 0.62	94.92 ± 0.61	0.8463 ± 0.0005	0.8449 ± 0.0005
Threshold, binary	94.98 ± 0.61	94.80 ± 0.62	0.8841 ± 0.0004	0.8835 ± 0.0004
Uninformative	94.94 ± 0.61	95.00 ± 0.60	1.0027 ± 0.0001	1.0014 ± 4.9e-05
Two-valued score	94.94 ± 0.61	95.02 ± 0.60	1.0011 ± 4.8e-05	1.000
Rare event	90.88 ± 0.80	90.80 ± 0.80	1.0027 ± 0.0001	1.0014 ± 4.8e-05

Table 2: Real prediction artifacts, 5,000 IID empirical-distribution repetitions per row. Widths and ratios show paired/mean 95% Monte Carlo error bars as appropriate; coverage errors are 1.96 binomial standard errors. Full method comparisons and Wilson intervals appear in Appendix F.

Artifact	n	PPI++ width	PPI++ cov. (%)	CF3 cov. (%)	CF3/PPI++ width
Galaxies	200	0.08598 ± 0.00016	94.46 ± 0.63	94.90 ± 0.61	1.0082 ± 0.0003
Galaxies	800	0.04299 ± 0.00004	95.08 ± 0.60	95.14 ± 0.60	1.0019 ± 0.0001
Forest	200	0.08527 ± 0.00019	94.38 ± 0.64	94.48 ± 0.63	1.007 ± 0.001
Forest	800	0.04269 ± 0.00005	94.66 ± 0.62	94.78 ± 0.62	0.9987 ± 0.0001
Protein disorder	200	0.06678 ± 0.00018	93.54 ± 0.68	94.50 ± 0.63	1.006 ± 0.001
Protein disorder	800	0.03331 ± 0.00004	94.92 ± 0.61	94.92 ± 0.61	1.0003 ± 0.0002

quadratic and threshold laws give smaller reductions, 15.4% and 11.6%, because irreducible binary noise and approximation error remain. The same comparisons to CF1 retain the nonlinear gains, so they are not explained by changing how a scalar coefficient is fit. Appendix B compares actual widths to the oracle values in Proposition 1.

For affine conditional means, CF3 offers no useful extra information and pays small estimation costs. For a two-valued score, CV3 and scalar PPI++ are numerically identical; CF3 and CF1 are also identical. In the null law, fitting coefficients cannot generate headroom. The real artifacts have very small cubic projection headroom (Appendix C), and Table 2 shows no practically substantial reduction. A cubic diagnostic cannot exclude richer nonlinear information beyond that span.

5.3 Narrower-looking intervals can be invalid

The full-sample CV3 plug-in approximation is fragile in the low-noise quadratic law: at $n = 200$, its coverage is 91.68 [90.88, 92.41]%, despite a narrow interval. CF3 gives 95.06 [94.42, 95.63]% in the same repetitions. Both are asymptotically justified at fixed dimension; these finite-sample results demonstrate why the theorem cannot be read as a guarantee at the nominal sample size. Shared estimation of moments and a small leading variance make the remainder consequential.

In the degree stress test, null-law degree-15 intervals at $n = 50$ have coverage 88.38 [87.46, 89.24]% and width ratio 0.935 ± 0.002 against degree one (Figure 1). Their apparent shortening is not an inferential gain. Other laws also show failure, even without consistent narrowing; all are retained in Appendix E. More flexible controls alone are not a safe recipe.

The rare-event law exposes a separate failure: with success probability 0.005, zero successes are common at $n = 200$. CF3 coverage is 63.46 [62.12, 64.78]%, and is still 90.80 [89.97, 91.57]% at $n = 800$. A zero empirical label variance can collapse any of the tuned score-based Wald intervals. The exact binomial and bounded-outcome intervals remain valid. The wider raw-score PPI interval sometimes masks the normal-approximation failure; this is not evidence that optimally tuned Wald intervals are uniformly valid.

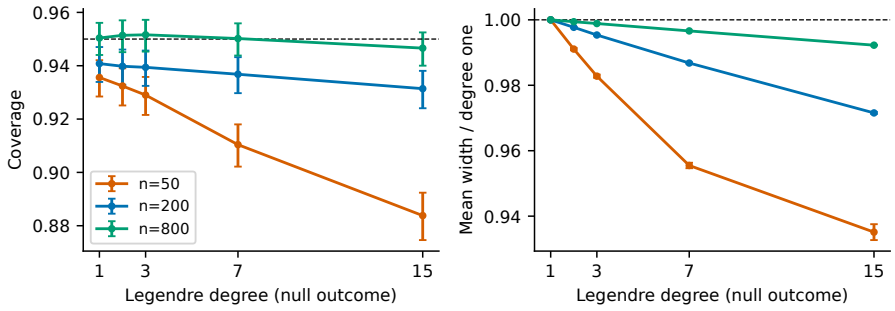


Figure 1: Degree stress test on uninformative scores. Five thousand repetitions per point. Left: coverage with Wilson 95% Monte Carlo intervals; right: width relative to degree one with paired errors. A shorter interval without nominal coverage is a failure.

Table 3: Finite-sample-valid intervals. Baseline is EB-label for continuous outcomes and the exact binomial interval for binary outcomes. At $n = 800$ there are 5,000 repetitions; at $12,800$ there are 3,000; $N = 10n$. Width bars are 95% Monte Carlo errors. Coverage is empirical, while the coverage lower bound follows from Theorem 1. Full Wilson intervals include the nonzero uncertainty when coverage is 100%.

Law	n	Baseline width	EB-split3 width	Cov. (%)	Width ratio
Quadratic, continuous	800	0.07565 ± 0.00003	0.09611 ± 0.00001	100.00 ± 0.00	1.2704 ± 0.0004
Quadratic, continuous	12800	$0.01411 \pm 2.1e\text{-}06$	$0.01012 \pm 5.1e\text{-}07$	100.00 ± 0.00	0.7169 ± 0.0001
Uninformative	800	$0.07039 \pm 1.7e\text{-}06$	0.20530 ± 0.00007	100.00 ± 0.00	2.917 ± 0.001
Uninformative	12800	$0.01740 \pm 3.3e\text{-}08$	$0.03639 \pm 4.4e\text{-}06$	99.97 ± 0.07	2.0914 ± 0.0003
Rare event	800	0.01110 ± 0.00007	0.05073 ± 0.00015	100.00 ± 0.00	4.569 ± 0.016
Rare event	12800	0.00252 ± 0.00001	0.00904 ± 0.00002	100.00 ± 0.00	3.581 ± 0.004

5.4 What survives a finite-sample guarantee?

The strict-coverage comparison changes the practical conclusion (Table 3). In the continuous quadratic example at $n = 800$, EB-split3 is wider than EB-label by the ratio 1.2704 ± 0.0004. At $n = 3,200$, the reduction is 6.1%; at $n = 12,800$, the ratio is 0.7169 ± 0.0001: a 28.3% reduction with coverage 100.00 [99.87, 100.00]%. This demonstrates a shorter interval with an actual finite-sample coverage theorem under fixed inputs. The large label count is a limitation, not an extra resource granted to the proposed method: every comparator in that row gets the same labels. EB-split3 remains much wider than an oracle-normal interval, so this result does not establish optimal finite-sample width.

In the uninformative and rare-event binary laws, exact binomial intervals are much narrower than the split bound. On the three real artifacts, even at $n = 800$ the strict split bound is wider than the existing valid label-only methods; the available results do not support practical strict-coverage improvement there. An unreported reduction against only EB-raw would be a weak comparison because ignoring the predictor already helps. Finally, the unlabeled-size check follows the ρ factor in the headroom formula: nonlinear gains increase when unlabeled observations are plentiful, but vanish in the no-information example. Figure 3 in the appendix reports coverage as well as width.

6 Discussion and limitations

With fixed observations and a fixed predictor, inference can still exploit nonlinear score information that scalar power tuning discards. The achievable first-order gain is limited by conditional noise and by estimating the score distribution. When the conditional mean is already affine, the regular score-only bound leaves no extra first-order gain. A low-dimensional dictionary can approximate a useful nonlinearity; it cannot guarantee improvement, and flexible empirical fitting can make intervals misleadingly small.

Our strongest numerical reductions concern asymptotic intervals in deliberately favorable synthetic laws. Strict finite-sample improvement is demonstrated only against specified bounded-outcome

baselines in a large-label continuous example. We do not show an improvement over the best possible finite-sample procedure. Exact binomial intervals decisively outperform our strict bound on binary no-information examples. We do not recommend replacing exact binary intervals with the split construction merely to use a predictor.

The real-data study conditions on empirical artifact distributions, and does not validate the original model's external training independence, annotations, spatial dependence, or deployment sampling. The theory needs representative IID samples from the same score law. Covariate shift, informative label selection, clustered observations, label noise relative to a scientific gold standard, or a predictor trained on the inference labels invalidate this unmodified analysis. We do not repair those violations. We study only means, bounded outcomes for exact guarantees, small prespecified dictionaries, and fixed folds. Gaussian asymptotics are pointwise rather than uniform near rare-event or degenerate distributions. Further transformations, extra recorded features, and other interval constructions might improve the small headroom found here; the cubic real-data result is not an impossibility theorem.

The reproducibility package contains all 233,000 reported repetitions in 49 conditions, including negative results, exact commands, environment versions, and access/failure logs. CPU resources and measured runtimes are in Appendix G. No human-subject data were collected. A practical benefit is more precise inference without additional annotation; a practical harm is false certainty if a narrow interval's assumptions or empirical coverage are ignored. Coverage failures are therefore first-class outputs. The finite-sample bounds are often conservative; equal nominal coverage does not imply equal achieved coverage. Existing artifact licenses are incompletely specified; code licensing is not treated as permission to relicense the data. This report was designed, implemented, and written by an autonomous language-model agent; the artifact enables independent review of its numerical and mathematical claims, which have not undergone human peer review.

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A Proofs and precise scope

A.1 Projection calculation and nonlinear headroom

Let all random variables be centered when computing covariance. Expand Equation (1):

$$nV(b) = \operatorname{Var} Y - 2b^\top c + (1 + n/N)b^\top \Sigma b. \tag{10}$$

Since $1+n/N = 1/\rho$, square completion gives Equations (2)–(3). The centered projection of m onto the dictionary is $c^\top \Sigma^{-1}(\phi - \mathbb{E}\phi)$ because $\operatorname{Cov}(\phi, Y) = \operatorname{Cov}(\phi, m)$. Its squared norm is $c^\top \Sigma^{-1} c$. For nested subspaces, orthogonality makes the difference in squared projection norms the squared norm of the added component. This proves the nested-gain statement.

For an arbitrary square-integrable control $h(S)$, $\operatorname{Cov}(Y, h) = \operatorname{Cov}(m, h)$. Complete the square in the Hilbert space of centered functions of S: the optimum is $h^*(S) = \rho\{m(S) - \mu\}$, up to an irrelevant constant, and $nV_S^* = \operatorname{Var} Y - \rho \operatorname{Var} m$. Apply the variance decomposition $\operatorname{Var} Y = \sigma_\epsilon^2 + \sigma_m^2$ and $1 - \rho = n/(n+N)$ to get Equation (4). Its last difference is the squared residual after affine projection. With a singular dictionary, perform these calculations on a fixed orthonormal basis for its nonconstant span. The pseudoinverse gives the same estimator whenever this span has fixed rank; this is the two-valued-score case in our implementation. We do not assert stable pseudoinverse behavior for an arbitrarily changing numerical rank.

A.2 Influence-function lower bound

For clarity, the regularity class in Proposition 2 consists of centered, finite-variance influence functions $a(Y, S)$ and $b(S)$ in an expansion

$$\widehat{\mu} - \mu = \frac{1}{n}\sum_{i=1}^{n} a(Y_i, S_i) + \frac{1}{N}\sum_{j=1}^{N} b(\widetilde{S}_j) + o_P(n^{-1/2}), \tag{11}$$

regular along all bounded, mean-zero likelihood-score paths $h(Y, S)$ of the unrestricted joint distribution. Such a path induces marginal score $\mathbb{E}[h \mid S]$ in the unlabeled sample. Differentiating the mean functional and the regular expansion gives

$$\mathbb{E}[ah] + \mathbb{E}[b\mathbb{E}(h \mid S)] = \mathbb{E}[(Y - \mu)h] \quad \text{for every such } h. \tag{12}$$

By iterated expectations and density of bounded centered scores in $L_0^2(P)$, $a(Y, S) + b(S) = Y - \mu$ almost surely; both sides have zero expectation. The first-order variance of Equation (11) is therefore

$$\frac{\mathbb{E}[(Y - \mu - b(S))^2]}{n} + \frac{\mathbb{E}[b(S)^2]}{N} = \frac{\sigma_\epsilon^2}{n} + \frac{\mathbb{E}[(m - \mu - b(S))^2]}{n} + \frac{\mathbb{E}[b(S)^2]}{N} \tag{13}$$

$$= V_S^* + (n^{-1} + N^{-1})\mathbb{E}[(b - \rho(m - \mu))^2]. \tag{14}$$

This proves the claimed lower bound and its equality condition. The argument is local and first-order. It does not exclude superefficiency at isolated distributions or shorter intervals with nonstandard coverage/length tradeoffs, and it is not a lower bound over arbitrary finite-sample confidence sets. Statements about oracle interval ratios in this paper concern normal, symmetric, first-order intervals only.

A.3 Consistency and Wald coverage

Let d be fixed. Under the fourth-moment and nonsingularity assumptions of Proposition 3, sample covariance matrices and cross-covariances converge to their population values. Since ρ converges to a constant, $\widehat{b} - b_d^* = o_P(1)$. Meanwhile, $\bar{\phi}_U - \bar{\phi}_L = O_P(n^{-1/2})$ when $n/N \to r \in (0, \infty)$. Thus

$$\widehat{\mu}_{\widehat{b}} - \widehat{\mu}_{b_d^*} = (\widehat{b} - b_d^*)^\top(\bar{\phi}_U - \bar{\phi}_L) = o_P(n^{-1/2}). \tag{15}$$

The two independent sample means for fixed b_d^* satisfy the multivariate central limit theorem; with positive limiting variance, the standardized oracle estimator tends to a standard normal distribution. Continuous mapping and the moment laws imply consistency of the plug-in variance, so Slutsky's theorem yields the Wald coverage statement.

In CF3, each fixed fold's training OLS slope converges to $\Sigma^{-1}c$, whether or not the conditional mean is cubic. Multiplying by ρ therefore gives $b_d^* + o_P(1)$. Fold proportions are fixed and positive, and each difference $\bar{\phi}_U - \bar{\phi}_{L_k}$ is $O_P(n^{-1/2})$. Summing the finitely many products proves asymptotic equivalence to the same oracle estimator. Within-fold sample residual variances consistently estimate $\mathrm{Var}(Y - b_d^{*\top}\phi)$. The average slope converges to b_d^*, so Equation (6) consistently estimates the oracle two-sample variance. All reported fold sizes are equal. With unequal fold sizes, Equation (5) and the average coefficient must instead use weights $|L_k|/n$, which the supplied implementation does.

These arguments explain fixed-dimensional asymptotic validity, but do not bound remainders at a particular n or for growing d. In particular, a small oracle variance can make an $o_P(n^{-1/2})$ term practically important. Neither the degree stress test nor rare-event failures contradict this pointwise theorem.

A.4 Finite-sample coverage and pilot cost

For a variable $Z \in [a, a+R]$, apply Maurer and Pontil [2009, Theorem 4] to $(Z-a)/R$ with failure probability $\delta/2$ and to $1-(Z-a)/R$ with the same probability. Scaling back gives Equation (7) simultaneously for both signs with total error at most δ. The theorem uses the unbiased sample variance, which equals the pairwise-difference definition in that source.

Let $\mathcal{P}$ be the pilot sigma field. The pilot is chosen by fixed indices before observing data. Conditional on $\mathcal{P}$, g is fixed, $0 \le g \le \rho_\ell$, and the inference pairs and unlabeled scores are independent IID samples from their original laws. The residual lies in $[-\rho_\ell, 1]$. Consequently,

$$\Pr\{|\overline{Y-g}_I - \mathbb{E}(Y-g)| > B_\ell(s_I^2(Y-g), 1+\rho_\ell, \alpha/2) \mid \mathcal{P}\} \le \alpha/2, \tag{16}$$

$$\Pr\{|\bar{g}_U - \mathbb{E}g| > B_N(s_U^2(g), \rho_\ell, \alpha/2) \mid \mathcal{P}\} \le \alpha/2. \tag{17}$$

On the complement of these two failure events, their sum differs from μ by at most Equation (8). A union bound and expectation over $\mathcal{P}$ give coverage at least $1-\alpha$. Clipping interval endpoints into $[0,1]$ cannot remove a true parameter in that set. No pilot calibration accuracy, conditional-mean specification, or large-sample approximation is needed.

To derive Equation (9), condition on $\mathcal{P}$ and expand

$$\mathrm{Var}(\overline{Y-g}_I + \bar{g}_U \mid \mathcal{P}) = \frac{\mathrm{Var}(Y-g)}{\ell} + \frac{\mathrm{Var}\, g}{N}. \tag{18}$$

The conditional mean is always μ. Square completion with respect to the centered control g gives Equation (9), up to constants in g that do not affect variance. Thus the unconditional estimator variance is the expectation of that expression. Relative to a full-data score oracle, its excess is

$$\begin{aligned}\sigma_\epsilon^2(\ell^{-1} - n^{-1}) + \sigma_m^2\{(\ell+N)^{-1} - (n+N)^{-1}\} \\ + (\ell^{-1} + N^{-1})\mathbb{E}_{\mathcal{P}}\,\mathrm{Var}\{g(S) - \rho_\ell m(S)\}.\end{aligned} \tag{19}$$

Every term is nonnegative. To outperform a full-data scalar oracle *in point-estimator variance*, the nonlinear headroom in Equation (4) must exceed these costs. To outperform its confidence-interval width, one must additionally account for the different coverage construction, including the explicit range penalties. This distinction is central to the finite-sample experiment.

The exact binary baseline uses $k = \sum_i Y_i$ and endpoints 0 if $k = 0$ (otherwise $\mathrm{Beta}^{-1}_{\alpha/2}(k, n-k+1)$), and 1 if $k = n$ (otherwise $\mathrm{Beta}^{-1}_{1-\alpha/2}(k+1, n-k)$). Inverting each binomial tail gives tail noncoverage at most $\alpha/2$; summing the two errors gives coverage at least $1-\alpha$. Binary support is known from the law or artifact, not inferred from whether the observed sample happens to contain only zeros and ones. Hoeffding's baseline has half-width $\sqrt{\log(2/\alpha)/(2n)}$ and the same endpoint clipping.

B Oracle checks and sample-size sensitivity

Table 4: Oracle width ratios and measured CF3 width at $n = 800$, $N = 8,000$. All-score ratios are first-order lower bounds under the regularity conditions of Proposition 2, not finite-sample interval bounds. Oracle moments are computed by exact piecewise-polynomial integration in `experiment.py`; measured widths use the actual repetitions.

Law	Cubic / linear oracle	All-score / linear bound	Cubic oracle width	CF3 width
Linear, continuous	1.000	1.000	0.00994	0.00994 ± 1.4e-06
Quadratic, continuous	0.309	0.309	0.01025	0.01025 ± 1.7e-06
Linear, binary	1.000	1.000	0.06019	0.06026 ± 0.00003
Quadratic, binary	0.844	0.844	0.05579	0.05590 ± 0.00003
Threshold, binary	0.883	0.724	0.04241	0.04246 ± 0.00004
Uninformative	1.000	1.000	0.06930	0.06944 ± 0.00001
Two-valued score	1.000	1.000	0.04481	0.04482 ± 0.00005
Rare event	1.000	1.000	0.00978	0.00942 ± 0.00007

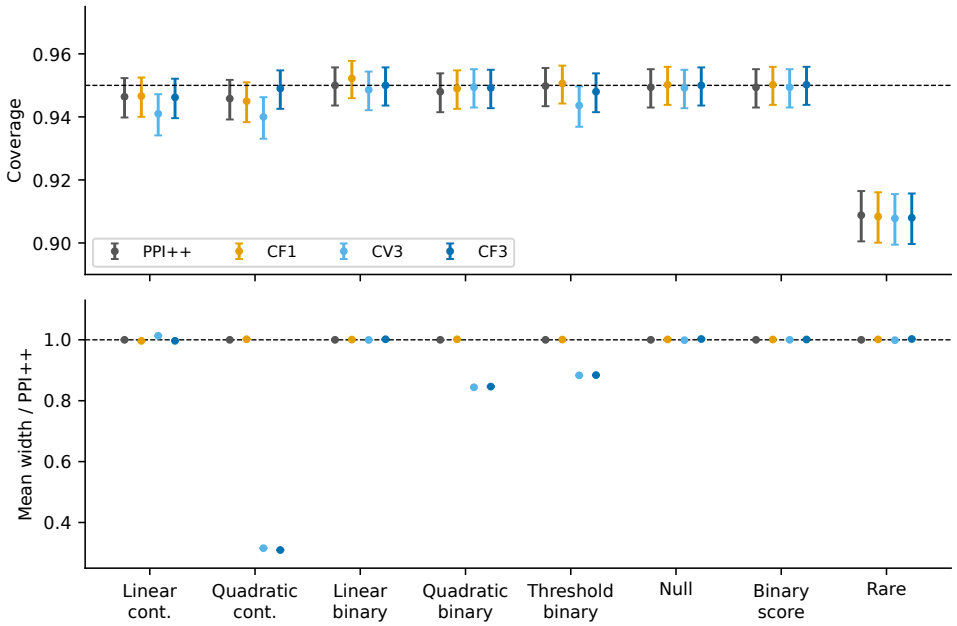


Figure 2: Main synthetic comparison. Five thousand repetitions per point; coverage bars are Wilson 95% intervals, and width-ratio bars are paired delta-method 95% errors. The rare-event failure remains visible. CV3's extra narrowing can be accompanied by undercoverage.

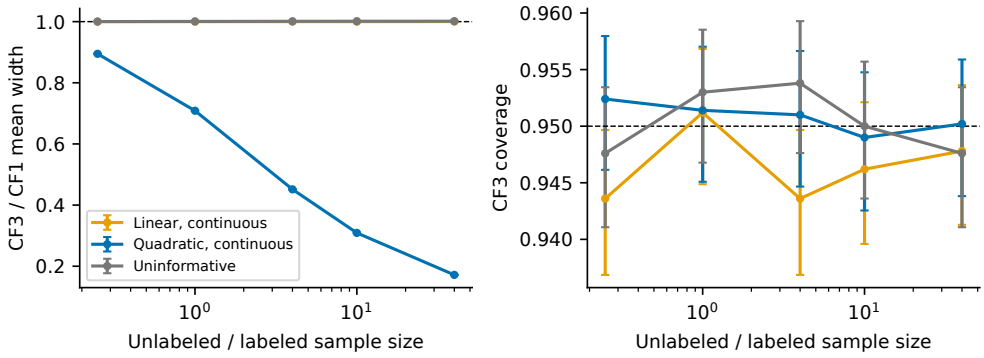


Figure 3: Unlabeled-size sensitivity with $n = 800$ and 5,000 repetitions per point. Width ratios compare CF3 against CF1, avoiding changes in the scalar fitting algorithm. Coverage bars are Wilson intervals.

C Prediction artifacts and data provenance

Table 5: Full artifacts define the empirical evaluation populations. The projection ratio is computed using artifact labels solely as an oracle diagnostic. The interval procedures receive only their sampled labeled pairs and sampled unlabeled scores.

Artifact	Records	Empirical target	Cubic/linear oracle width
Galaxies	16743	0.259273	1.0000
Forest	1596	0.151629	0.9961
Protein disorder	10802	0.177375	0.9987

The artifact version is pinned to the Google Drive identifiers in the `v0.2.3` loader at `https://github.com/aangelopoulos/ppi_py/tree/v0.2.3`. The galaxies notebook attributes labels to Galaxy Zoo 2, based on Sloan Digital Sky Survey images, and describes the spiral-arm prevalence target. The forest notebook attributes field labels to Bullock and colleagues and tree-cover predictions to Sexton and colleagues. The alphafold notebook attributes disorder-related labels to Iakoucheva and colleagues and predicted structures/disorder processing to Jumper and colleagues and Bludau and colleagues. These attributions were read in the version-pinned notebooks, which are retained in `literature/`; the original source studies were not used for additional empirical claims or added to the reference list without being read. Our alphafold target is mean disorder prevalence, rather than the notebook's modification odds ratio.

The reference code is MIT licensed, with its LICENSE retained. No independent license for the downloaded NPZ files is supplied in the repository materials checked here. We record that uncertainty rather than assign them the code license. Download instructions, source identifiers, shapes, checksums, and access dates are in the artifact. No new human annotations, personally identifying records, or new model checkpoints are released. Files in the task workspace are research inputs; downstream redistribution should resolve the source data terms independently.

Sampling with replacement can select the same source row more than once, including on both sides of the labeled/unlabeled split. The *indices are independent draws*, so the random variables are independent under the specified empirical-distribution experiment, despite coincident row values. This is different from pretending that a without-replacement partition is an IID superpopulation sample, and different from obtaining new annotations for duplicated records. It estimates resampling performance on the artifact distribution, not the number of unique physical items a scientific study would annotate.

D Additional finite-sample results

Figure 4 and the following table include intermediate sample sizes and every strict-coverage comparator. "Baseline" in the figure means EB-label for the continuous law and exact binomial for the binary laws. Pointwise Wilson intervals should be used to interpret empirical coverage equal to one: it is not proof of population coverage exactly one.

Table 6: All finite-sample comparators; widths with 95% Monte Carlo errors, coverage with Wilson 95% intervals, and paired width ratios.

Law	n	Method	Width	Coverage (%)	Width / EB-label
		Hoeffding	0.19206 ± 0.00000	100.00 [99.92, 100.00]	0.9473 ± 0.0005
Quadratic, continuous	200	EB-label	0.20275 ± 0.00011	100.00 [99.92, 100.00]	1.000
		EB-raw	0.45960 ± 0.00025	100.00 [99.92, 100.00]	2.267 ± 0.001
		EB-split1	0.42551 ± 0.00025	100.00 [99.92, 100.00]	2.099 ± 0.001
		EB-split3	0.34196 ± 0.00003	100.00 [99.92, 100.00]	1.687 ± 0.001
		Hoeffding	0.09603 ± 0.00000	100.00 [99.92, 100.00]	1.2694 ± 0.0004
Quadratic, continuous	800	EB-label	0.07565 ± 0.00003	99.98 [99.89, 100.00]	1.000
		EB-raw	0.16722 ± 0.00006	100.00 [99.92, 100.00]	2.210 ± 0.001
		EB-split1	0.13587 ± 0.00005	100.00 [99.92, 100.00]	1.796 ± 0.001
		EB-split3	0.09611 ± 0.00001	100.00 [99.92, 100.00]	1.2704 ± 0.0004
		Hoeffding	0.04802 ± 0.00000	100.00 [99.87, 100.00]	1.5279 ± 0.0004
Quadratic, continuous	3200	EB-label	0.03143 ± 0.00001	99.97 [99.81, 99.99]	1.000

Law	n	Method	Width	Coverage (%)	Width / EB-label
		EB-raw	0.06805 ± 0.00002	100.00 [99.87, 100.00]	2.165 ± 0.001
		EB-split1	0.04897 ± 0.00002	100.00 [99.87, 100.00]	1.5583 ± 0.0004
		EB-split3	0.02950 ± 2.0e-06	100.00 [99.87, 100.00]	0.9386 ± 0.0002
		Hoeffding	0.02401 ± 0.00000	100.00 [99.87, 100.00]	1.7010 ± 0.0003
Quadratic, continuous	12800	EB-label	0.01411 ± 2.1e-06	99.93 [99.76, 99.98]	1.000
		EB-raw	0.03014 ± 0.00001	100.00 [99.87, 100.00]	2.1356 ± 0.0003
		EB-split1	0.01977 ± 3.7e-06	100.00 [99.87, 100.00]	1.4007 ± 0.0002
		EB-split3	0.01012 ± 5.1e-07	100.00 [99.87, 100.00]	0.7169 ± 0.0001
		Hoeffding	0.19206 ± 0.00000	99.44 [99.19, 99.61]	0.6154 ± 4.1e-05
		Exact-binomial	0.14234 ± 0.00001	95.48 [94.87, 96.02]	0.4561 ± 1.3e-05
Uninformative	200	EB-label	0.31209 ± 0.00002	100.00 [99.92, 100.00]	1.000
		EB-raw	0.55086 ± 0.00023	100.00 [99.92, 100.00]	1.765 ± 0.001
		EB-split1	0.56099 ± 0.00029	100.00 [99.92, 100.00]	1.798 ± 0.001
		EB-split3	0.57585 ± 0.00039	100.00 [99.92, 100.00]	1.845 ± 0.001
		Hoeffding	0.09603 ± 0.00000	99.28 [99.00, 99.48]	0.7372 ± 1.4e-05
		Exact-binomial	0.07039 ± 1.7e-06	95.70 [95.10, 96.23]	0.5404 ± 2.3e-06
Uninformative	800	EB-label	0.13026 ± 2.6e-06	99.96 [99.85, 99.99]	1.000
		EB-raw	0.21290 ± 0.00006	100.00 [99.92, 100.00]	1.6344 ± 0.0004
		EB-split1	0.20247 ± 0.00006	100.00 [99.92, 100.00]	1.5543 ± 0.0004
		EB-split3	0.20530 ± 0.00007	100.00 [99.92, 100.00]	1.576 ± 0.001
		Hoeffding	0.04802 ± 0.00000	99.53 [99.22, 99.72]	0.8176 ± 5.6e-06
		Exact-binomial	0.03494 ± 2.6e-07	95.10 [94.27, 95.82]	0.5950 ± 4.5e-07
Uninformative	3200	EB-label	0.05873 ± 4.0e-07	99.97 [99.81, 99.99]	1.000
		EB-raw	0.09087 ± 0.00002	100.00 [99.87, 100.00]	1.5474 ± 0.0003
		EB-split1	0.08203 ± 0.00002	100.00 [99.87, 100.00]	1.3969 ± 0.0003
		EB-split3	0.08266 ± 0.00002	100.00 [99.87, 100.00]	1.4075 ± 0.0003
		Hoeffding	0.02401 ± 0.00000	99.57 [99.26, 99.75]	0.8647 ± 1.5e-06
		Exact-binomial	0.01740 ± 3.3e-08	94.97 [94.13, 95.69]	0.6267 ± 6.3e-08
Uninformative	12800	EB-label	0.02776 ± 5.0e-08	99.90 [99.71, 99.97]	1.000
		EB-raw	0.04154 ± 4.6e-06	100.00 [99.87, 100.00]	1.4962 ± 0.0002
		EB-split1	0.03625 ± 3.8e-06	99.97 [99.81, 99.99]	1.3056 ± 0.0001
		EB-split3	0.03639 ± 4.4e-06	99.97 [99.81, 99.99]	1.3106 ± 0.0002
		Hoeffding	0.10103 ± 0.00014	100.00 [99.92, 100.00]	1.489 ± 0.007
		Exact-binomial	0.02648 ± 0.00021	97.94 [97.51, 98.30]	0.390 ± 0.001
Rare event	200	EB-label	0.06783 ± 0.00039	100.00 [99.92, 100.00]	1.000
		EB-raw	0.21687 ± 0.00062	100.00 [99.92, 100.00]	3.197 ± 0.018
		EB-split1	0.16750 ± 0.00048	100.00 [99.92, 100.00]	2.469 ± 0.008
		EB-split3	0.16821 ± 0.00049	100.00 [99.92, 100.00]	2.480 ± 0.008
		Hoeffding	0.05302 ± 0.00007	100.00 [99.92, 100.00]	2.129 ± 0.008
		Exact-binomial	0.01110 ± 0.00007	95.94 [95.36, 96.45]	0.4459 ± 0.0005
Rare event	800	EB-label	0.02490 ± 0.00012	100.00 [99.92, 100.00]	1.000
		EB-raw	0.08004 ± 0.00031	100.00 [99.92, 100.00]	3.214 ± 0.017
		EB-split1	0.05058 ± 0.00015	100.00 [99.92, 100.00]	2.031 ± 0.005
		EB-split3	0.05073 ± 0.00015	100.00 [99.92, 100.00]	2.037 ± 0.005
		Hoeffding	0.02900 ± 0.00005	100.00 [99.87, 100.00]	2.449 ± 0.009
		Exact-binomial	0.00520 ± 0.00002	95.03 [94.20, 95.75]	0.4395 ± 0.0004
Rare event	3200	EB-label	0.01184 ± 0.00006	100.00 [99.87, 100.00]	1.000
		EB-raw	0.03451 ± 0.00020	100.00 [99.87, 100.00]	2.914 ± 0.020
		EB-split1	0.01869 ± 0.00007	100.00 [99.87, 100.00]	1.579 ± 0.004
		EB-split3	0.01873 ± 0.00007	100.00 [99.87, 100.00]	1.582 ± 0.004
		Hoeffding	0.01701 ± 0.00002	100.00 [99.87, 100.00]	3.219 ± 0.001
		Exact-binomial	0.00252 ± 0.00001	95.23 [94.41, 95.94]	0.4777 ± 0.0003
Rare event	12800	EB-label	0.00528 ± 0.00001	100.00 [99.87, 100.00]	1.000
		EB-raw	0.01780 ± 0.00010	100.00 [99.87, 100.00]	3.370 ± 0.018
		EB-split1	0.00902 ± 0.00001	100.00 [99.87, 100.00]	1.708 ± 0.001
		EB-split3	0.00904 ± 0.00002	100.00 [99.87, 100.00]	1.711 ± 0.001

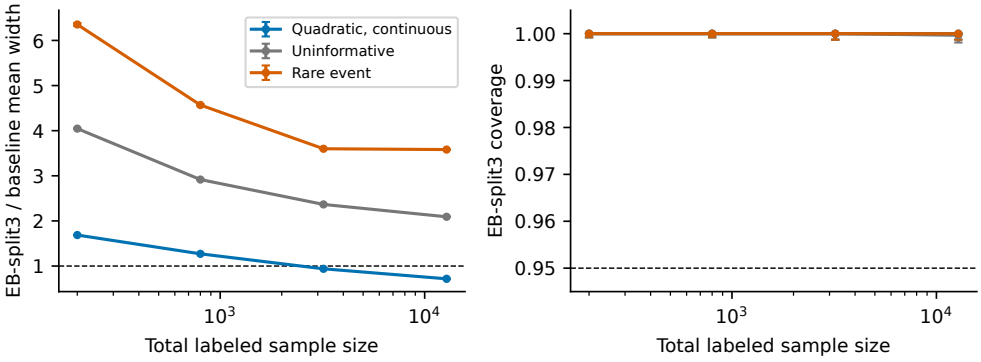


Figure 4: Strict-coverage tradeoff as the label count changes; $N = 10n$. All methods within a point have identical inputs. Error bars are Wilson intervals for coverage and paired delta-method errors for ratios.

E Dictionary degree stress test

All degrees in this test use full-sample pooled-moment controls, not CF3. The test is a deliberate stress test of an asymptotic approximation as the dictionary becomes large relative to the labels. Each cell has 5,000 paired repetitions, independent from the main suite through separate deterministic seeds. Width reduction without coverage is not counted as a success. Table errors are 95% Monte Carlo errors; coverage uses Wilson intervals.

Table 7: All degree stress-test results; 5,000 repetitions per condition. Coverage intervals are Wilson 95% Monte Carlo intervals.

Law	n	Degree	Width	Coverage (%)	Width / degree one
Linear, continuous	50	1	0.04192 ± 0.00009	94.46 [93.79, 95.06]	1.000
		2	0.04565 ± 0.00017	92.06 [91.28, 92.78]	1.089 ± 0.003
		3	0.04900 ± 0.00027	91.02 [90.20, 91.78]	1.169 ± 0.005
		7	0.06349 ± 0.00061	89.30 [88.41, 90.13]	1.514 ± 0.014
		15	0.09510 ± 0.00122	90.66 [89.82, 91.44]	2.269 ± 0.028
Linear, continuous	200	1	0.02015 ± 0.00001	94.62 [93.96, 95.21]	1.000
		2	0.02066 ± 0.00002	93.76 [93.06, 94.40]	1.026 ± 0.001
		3	0.02114 ± 0.00003	93.12 [92.38, 93.79]	1.049 ± 0.001
		7	0.02308 ± 0.00007	91.58 [90.78, 92.32]	1.145 ± 0.003
		15	0.02700 ± 0.00014	89.04 [88.14, 89.88]	1.340 ± 0.007
Linear, continuous	800	1	0.00998 ± 2.0e-06	94.94 [94.30, 95.51]	1.000
		2	0.01004 ± 3.3e-06	94.88 [94.23, 95.46]	1.0066 ± 0.0003
		3	0.01011 ± 4.5e-06	94.80 [94.15, 95.38]	1.0130 ± 0.0004
		7	0.01036 ± 0.00001	93.52 [92.80, 94.17]	1.038 ± 0.001
		15	0.01087 ± 0.00001	92.54 [91.78, 93.24]	1.089 ± 0.001
Quadratic, continuous	50	1	0.13005 ± 0.00030	92.76 [92.01, 93.45]	1.000
		2	0.04777 ± 0.00019	89.88 [89.01, 90.69]	0.367 ± 0.002
		3	0.05188 ± 0.00033	87.86 [86.93, 88.74]	0.399 ± 0.002
		7	0.06791 ± 0.00077	79.24 [78.09, 80.34]	0.522 ± 0.006
		15	0.10377 ± 0.00155	71.46 [70.19, 72.69]	0.798 ± 0.011
Quadratic, continuous	200	1	0.06603 ± 0.00007	94.16 [93.48, 94.78]	1.000
		2	0.02143 ± 0.00003	93.64 [92.93, 94.28]	0.3246 ± 0.0005
		3	0.02204 ± 0.00005	92.32 [91.55, 93.03]	0.334 ± 0.001
		7	0.02439 ± 0.00010	85.88 [84.89, 86.82]	0.369 ± 0.001
		15	0.02902 ± 0.00019	72.68 [71.43, 73.90]	0.440 ± 0.003
Quadratic, continuous	800	1	0.03311 ± 0.00002	94.66 [94.00, 95.25]	1.000
		2	0.01037 ± 3.7e-06	94.60 [93.94, 95.19]	0.3132 ± 0.0002
		3	0.01045 ± 0.00001	93.78 [93.08, 94.42]	0.3156 ± 0.0002
		7	0.01075 ± 0.00001	90.76 [89.93, 91.53]	0.3247 ± 0.0003
		15	0.01136 ± 0.00002	83.76 [82.71, 84.76]	0.343 ± 0.001
Uninformative	50	1	0.27463 ± 0.00015	93.56 [92.85, 94.21]	1.000
		2	0.27218 ± 0.00018	93.24 [92.51, 93.90]	0.9911 ± 0.0004
		3	0.26991 ± 0.00020	92.90 [92.15, 93.58]	0.9828 ± 0.0005
		7	0.26241 ± 0.00026	91.04 [90.22, 91.80]	0.955 ± 0.001
		15	0.25681 ± 0.00067	88.38 [87.46, 89.24]	0.935 ± 0.002
Uninformative	200	1	0.13827 ± 0.00002	94.08 [93.39, 94.70]	1.000
		2	0.13795 ± 0.00002	93.98 [93.29, 94.61]	0.9977 ± 0.0001
		3	0.13763 ± 0.00003	93.94 [93.24, 94.57]	0.9954 ± 0.0001
		7	0.13644 ± 0.00003	93.68 [92.97, 94.32]	0.9868 ± 0.0002
		15	0.13433 ± 0.00005	93.14 [92.41, 93.81]	0.9715 ± 0.0003
Uninformative	800	1	0.06926 ± 2.3e-06	95.04 [94.40, 95.61]	1.000
		2	0.06922 ± 2.8e-06	95.14 [94.51, 95.70]	0.9994 ± 2.3e-05
		3	0.06918 ± 3.2e-06	95.16 [94.53, 95.72]	0.9989 ± 3.1e-05
		7	0.06902 ± 4.4e-06	95.02 [94.38, 95.59]	0.9966 ± 0.0001
		15	0.06872 ± 0.00001	94.66 [94.00, 95.25]	0.9923 ± 0.0001

F Full main results

All tables below use $N = 10n$ and 5,000 repetitions. Widths show mean ± 1.96 standard errors; coverage shows the estimate followed by its Wilson 95% Monte Carlo interval, in percent. Ratios compare mean widths to unrestricted scalar PPI++ with paired delta-method errors. They are descriptive ratios, including when either procedure under-covers. Exact-binomial rows are present only for known binary outcomes. The clipped PPI++ rows use the reference package's $[0, 1]$ coefficient restriction; all methods use the declared unbiased covariance/variance convention. Large Monte Carlo errors or zero mean widths are never silently removed.

F.1 Linear continuous

Table 8: Linear continuous, $n = 200$, $N = 2000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.06416 ± 0.00006	94.62 [93.96, 95.21]	3.184 ± 0.003
PPI	0.03032 ± 0.00001	95.04 [94.40, 95.61]	1.504 ± 0.001
PPI++	0.02015 ± 0.00001	94.62 [93.96, 95.21]	1.000
PPI++clip	0.02015 ± 0.00001	94.62 [93.96, 95.21]	1.000
CF1	0.01989 ± 0.00001	95.04 [94.40, 95.61]	0.987 ± 0.001
CV3	0.02116 ± 0.00004	93.14 [92.41, 93.81]	1.050 ± 0.002
CF3	0.01990 ± 0.00001	94.96 [94.32, 95.53]	0.987 ± 0.001
Hoeffding	0.19206 ± 0.00000	100.00 [99.92, 100.00]	9.531 ± 0.006
EB-label	0.19968 ± 0.00009	100.00 [99.92, 100.00]	9.908 ± 0.006
EB-raw	0.31814 ± 0.00003	100.00 [99.92, 100.00]	15.787 ± 0.009
EB-split1	0.34059 ± 0.00002	100.00 [99.92, 100.00]	16.901 ± 0.010
EB-split3	0.34075 ± 0.00003	100.00 [99.92, 100.00]	16.909 ± 0.010

Table 9: Linear continuous, $n = 800$, $N = 8000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.03209 ± 0.00001	95.20 [94.57, 95.76]	3.217 ± 0.001
PPI	0.01516 ± 2.6e-06	94.04 [93.35, 94.66]	1.5194 ± 0.0003
PPI++	0.00998 ± 2.0e-06	94.64 [93.98, 95.23]	1.000
PPI++clip	0.00998 ± 2.0e-06	94.64 [93.98, 95.23]	1.000
CF1	0.00994 ± 1.4e-06	94.66 [94.00, 95.25]	0.9966 ± 0.0001
CV3	0.01011 ± 4.7e-06	94.10 [93.41, 94.72]	1.0133 ± 0.0004
CF3	0.00994 ± 1.4e-06	94.62 [93.96, 95.21]	0.9967 ± 0.0001
Hoeffding	0.09603 ± 0.00000	100.00 [99.92, 100.00]	9.625 ± 0.002
EB-label	0.07407 ± 0.00002	100.00 [99.92, 100.00]	7.424 ± 0.002
EB-raw	0.09639 ± 0.00001	100.00 [99.92, 100.00]	9.660 ± 0.002
EB-split1	0.09550 ± 0.00001	100.00 [99.92, 100.00]	9.571 ± 0.002
EB-split3	0.09551 ± 0.00001	100.00 [99.92, 100.00]	9.573 ± 0.002

F.2 Quadratic continuous

Table 10: Quadratic continuous, $n = 200$, $N = 2000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.06620 ± 0.00007	95.14 [94.51, 95.70]	1.0036 ± 0.0001
PPI	0.10677 ± 0.00015	94.58 [93.92, 95.17]	1.619 ± 0.002
PPI++	0.06596 ± 0.00007	94.92 [94.28, 95.49]	1.000
PPI++clip	0.06608 ± 0.00007	95.06 [94.42, 95.63]	1.0017 ± 0.0001
CF1	0.06651 ± 0.00007	95.18 [94.55, 95.74]	1.0083 ± 0.0003
CV3	0.02201 ± 0.00004	91.68 [90.88, 92.41]	0.334 ± 0.001
CF3	0.02051 ± 0.00001	95.06 [94.42, 95.63]	0.3109 ± 0.0003
Hoeffding	0.19206 ± 0.00000	100.00 [99.92, 100.00]	2.912 ± 0.003
EB-label	0.20275 ± 0.00011	100.00 [99.92, 100.00]	3.074 ± 0.002
EB-raw	0.45960 ± 0.00025	100.00 [99.92, 100.00]	6.967 ± 0.007
EB-split1	0.42551 ± 0.00025	100.00 [99.92, 100.00]	6.451 ± 0.006
EB-split3	0.34196 ± 0.00003	100.00 [99.92, 100.00]	5.184 ± 0.006

Table 11: Quadratic continuous, $n = 800$, $N = 8000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.03314 ± 0.00002	94.60 [93.94, 95.19]	1.0009 ± 3.6e-05
PPI	0.05345 ± 0.00004	95.02 [94.38, 95.59]	1.614 ± 0.001
PPI++	0.03311 ± 0.00002	94.58 [93.92, 95.17]	1.000
PPI++clip	0.03313 ± 0.00002	94.64 [93.98, 95.23]	1.0004 ± 2.8e-05
CF1	0.03318 ± 0.00002	94.50 [93.83, 95.10]	1.0020 ± 0.0001
CV3	0.01045 ± 0.00001	94.00 [93.31, 94.63]	0.3155 ± 0.0002
CF3	0.01025 ± 1.7e-06	94.90 [94.25, 95.48]	0.3096 ± 0.0002
Hoeffding	0.09603 ± 0.00000	100.00 [99.92, 100.00]	2.900 ± 0.002
EB-label	0.07565 ± 0.00003	99.98 [99.89, 100.00]	2.2848 ± 0.0004
EB-raw	0.16722 ± 0.00006	100.00 [99.92, 100.00]	5.050 ± 0.002
EB-split1	0.13587 ± 0.00005	100.00 [99.92, 100.00]	4.103 ± 0.002
EB-split3	0.09611 ± 0.00001	100.00 [99.92, 100.00]	2.903 ± 0.001

F.3 Linear binary

Table 12: Linear binary, $n = 200$, $N = 2000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.13858 ± 0.00001	94.36 [93.69, 94.97]	1.154 ± 0.001
PPI	0.12115 ± 0.00013	94.80 [94.15, 95.38]	1.0088 ± 0.0002
PPI++	0.12009 ± 0.00011	94.52 [93.85, 95.12]	1.000
PPI++clip	0.12009 ± 0.00011	94.50 [93.83, 95.10]	1.0000 ± 8.0e-06
CF1	0.12044 ± 0.00012	94.78 [94.13, 95.36]	1.0029 ± 0.0002
CV3	0.11990 ± 0.00011	94.34 [93.66, 94.95]	0.9984 ± 0.0001
CF3	0.12097 ± 0.00012	94.70 [94.04, 95.29]	1.0073 ± 0.0003
Hoeffding	0.19206 ± 0.00000	99.38 [99.12, 99.56]	1.599 ± 0.002
EB-label	0.31208 ± 0.00002	100.00 [99.92, 100.00]	2.599 ± 0.002
EB-raw	0.48358 ± 0.00022	100.00 [99.92, 100.00]	4.027 ± 0.002
EB-split1	0.54830 ± 0.00028	100.00 [99.92, 100.00]	4.566 ± 0.004
EB-split3	0.55405 ± 0.00033	100.00 [99.92, 100.00]	4.613 ± 0.004
Exact-binomial	0.14233 ± 0.00001	95.86 [95.27, 96.38]	1.185 ± 0.001

Table 13: Linear binary, $n = 800$, $N = 8000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.06929 ± 1.8e-06	94.44 [93.77, 95.04]	1.152 ± 0.001
PPI	0.06063 ± 0.00003	95.10 [94.47, 95.66]	1.0081 ± 0.0001
PPI++	0.06014 ± 0.00003	95.00 [94.36, 95.57]	1.000
PPI++clip	0.06014 ± 0.00003	95.00 [94.36, 95.57]	1.0000 ± 3.0e-08
CF1	0.06019 ± 0.00003	95.22 [94.59, 95.78]	1.0008 ± 0.0001
CV3	0.06012 ± 0.00003	94.86 [94.21, 95.44]	0.9996 ± 2.9e-05
CF3	0.06026 ± 0.00003	95.00 [94.36, 95.57]	1.0019 ± 0.0001
Hoeffding	0.09603 ± 0.00000	99.12 [98.82, 99.34]	1.597 ± 0.001
EB-label	0.13026 ± 2.7e-06	99.98 [99.89, 100.00]	2.166 ± 0.001
EB-raw	0.17920 ± 0.00006	100.00 [99.92, 100.00]	2.9795 ± 0.0005
EB-split1	0.19907 ± 0.00007	100.00 [99.92, 100.00]	3.310 ± 0.001
EB-split3	0.19967 ± 0.00007	100.00 [99.92, 100.00]	3.320 ± 0.001
Exact-binomial	0.07039 ± 1.8e-06	95.30 [94.68, 95.85]	1.170 ± 0.001

F.4 Quadratic binary

Table 14: Quadratic binary, $n = 200$, $N = 2000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.13216 ± 0.00008	94.46 [93.79, 95.06]	1.0032 ± 0.0001
PPI	0.15649 ± 0.00018	94.92 [94.28, 95.49]	1.188 ± 0.001
PPI++	0.13174 ± 0.00009	94.60 [93.94, 95.19]	1.000
PPI++clip	0.13195 ± 0.00008	94.56 [93.90, 95.16]	1.0016 ± 0.0001
CF1	0.13260 ± 0.00009	94.66 [94.00, 95.25]	1.0065 ± 0.0002
CV3	0.11113 ± 0.00013	93.96 [93.27, 94.59]	0.844 ± 0.001
CF3	0.11233 ± 0.00014	94.98 [94.34, 95.55]	0.853 ± 0.001
Hoeffding	0.19206 ± 0.00000	99.60 [99.38, 99.74]	1.458 ± 0.001
EB-label	0.30238 ± 0.00013	100.00 [99.92, 100.00]	2.295 ± 0.001
EB-raw	0.54187 ± 0.00031	100.00 [99.92, 100.00]	4.113 ± 0.003
EB-split1	0.55029 ± 0.00037	100.00 [99.92, 100.00]	4.177 ± 0.003
EB-split3	0.53944 ± 0.00035	100.00 [99.92, 100.00]	4.095 ± 0.003
Exact-binomial	0.13600 ± 0.00008	95.30 [94.68, 95.85]	1.0324 ± 0.0001

Table 15: Quadratic binary, $n = 800$, $N = 8000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.06610 ± 0.00002	94.54 [93.88, 95.14]	1.0007 ± 2.9e-05
PPI	0.07830 ± 0.00004	94.18 [93.50, 94.80]	1.185 ± 0.001
PPI++	0.06605 ± 0.00002	94.80 [94.15, 95.38]	1.000
PPI++clip	0.06608 ± 0.00002	94.62 [93.96, 95.21]	1.0004 ± 2.3e-05
CF1	0.06617 ± 0.00002	94.90 [94.25, 95.48]	1.0017 ± 0.0001
CV3	0.05575 ± 0.00003	94.94 [94.30, 95.51]	0.8440 ± 0.0005
CF3	0.05590 ± 0.00003	94.92 [94.28, 95.49]	0.8463 ± 0.0005
Hoeffding	0.09603 ± 0.00000	99.64 [99.43, 99.77]	1.4539 ± 0.0005
EB-label	0.12544 ± 0.00003	99.98 [99.89, 100.00]	1.8990 ± 0.0001
EB-raw	0.20841 ± 0.00007	100.00 [99.92, 100.00]	3.155 ± 0.001
EB-split1	0.19693 ± 0.00008	100.00 [99.92, 100.00]	2.981 ± 0.001
EB-split3	0.19217 ± 0.00008	100.00 [99.92, 100.00]	2.909 ± 0.001
Exact-binomial	0.06721 ± 0.00002	95.38 [94.76, 95.93]	1.0175 ± 3.1e-05

F.5 Threshold binary

Table 16: Threshold binary, $n = 200$, $N = 2000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.12912 ± 0.00010	95.20 [94.57, 95.76]	1.346 ± 0.002
PPI	0.09601 ± 0.00012	95.16 [94.53, 95.72]	1.0009 ± 0.0001
PPI++	0.09593 ± 0.00013	94.90 [94.25, 95.48]	1.000
PPI++clip	0.09601 ± 0.00012	94.98 [94.34, 95.55]	1.0009 ± 0.0001
CF1	0.09619 ± 0.00013	95.22 [94.59, 95.78]	1.0027 ± 0.0002
CV3	0.08496 ± 0.00015	94.40 [93.73, 95.00]	0.886 ± 0.001
CF3	0.08527 ± 0.00016	94.94 [94.30, 95.51]	0.889 ± 0.001
Hoeffding	0.19206 ± 0.00000	99.82 [99.66, 99.91]	2.002 ± 0.003
EB-label	0.29779 ± 0.00015	100.00 [99.92, 100.00]	3.104 ± 0.004
EB-raw	0.44155 ± 0.00021	100.00 [99.92, 100.00]	4.603 ± 0.004
EB-split1	0.49994 ± 0.00027	100.00 [99.92, 100.00]	5.212 ± 0.005
EB-split3	0.48884 ± 0.00036	100.00 [99.92, 100.00]	5.096 ± 0.005
Exact-binomial	0.13301 ± 0.00010	96.12 [95.55, 96.62]	1.387 ± 0.002

Table 17: Threshold binary, $n = 800$, $N = 8000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.06463 ± 0.00002	95.10 [94.47, 95.66]	1.346 ± 0.001
PPI	0.04805 ± 0.00003	95.08 [94.45, 95.65]	1.0005 ± 2.9e-05
PPI++	0.04802 ± 0.00003	94.98 [94.34, 95.55]	1.000
PPI++clip	0.04805 ± 0.00003	95.06 [94.42, 95.63]	1.0005 ± 2.8e-05
CF1	0.04806 ± 0.00003	95.06 [94.42, 95.63]	1.0007 ± 4.8e-05
CV3	0.04242 ± 0.00004	94.36 [93.69, 94.97]	0.8834 ± 0.0004
CF3	0.04246 ± 0.00004	94.80 [94.15, 95.38]	0.8841 ± 0.0004
Hoeffding	0.09603 ± 0.00000	99.70 [99.51, 99.82]	2.000 ± 0.001
EB-label	0.12321 ± 0.00004	100.00 [99.92, 100.00]	2.566 ± 0.002
EB-raw	0.15815 ± 0.00005	100.00 [99.92, 100.00]	3.293 ± 0.001
EB-split1	0.17510 ± 0.00007	100.00 [99.92, 100.00]	3.646 ± 0.001
EB-split3	0.16880 ± 0.00008	100.00 [99.92, 100.00]	3.515 ± 0.001
Exact-binomial	0.06574 ± 0.00002	95.46 [94.85, 96.00]	1.369 ± 0.001

F.6 Uninformative binary

Table 18: Uninformative binary, $n = 200$, $N = 2000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.13859 ± 0.00001	94.12 [93.43, 94.74]	1.0023 ± 0.0001
PPI	0.16185 ± 0.00014	94.34 [93.66, 94.95]	1.171 ± 0.001
PPI++	0.13826 ± 0.00002	94.28 [93.60, 94.89]	1.000
PPI++clip	0.13842 ± 0.00002	94.24 [93.56, 94.85]	1.0011 ± 0.0001
CF1	0.13899 ± 0.00004	94.42 [93.75, 95.02]	1.0052 ± 0.0002
CV3	0.13764 ± 0.00002	94.24 [93.56, 94.85]	0.9955 ± 0.0001
CF3	0.13978 ± 0.00004	94.56 [93.90, 95.16]	1.0109 ± 0.0003
Hoeffding	0.19206 ± 0.00000	99.44 [99.19, 99.61]	1.3891 ± 0.0002
EB-label	0.31209 ± 0.00002	100.00 [99.92, 100.00]	2.2572 ± 0.0002
EB-raw	0.55086 ± 0.00023	100.00 [99.92, 100.00]	3.984 ± 0.002
EB-split1	0.56099 ± 0.00029	100.00 [99.92, 100.00]	4.057 ± 0.002
EB-split3	0.57585 ± 0.00039	100.00 [99.92, 100.00]	4.165 ± 0.003
Exact-binomial	0.14234 ± 0.00001	95.48 [94.87, 96.02]	1.0295 ± 0.0001

Table 19: Uninformative binary, $n = 800$, $N = 8000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.06930 ± 1.7e-06	94.96 [94.32, 95.53]	1.0006 ± 2.2e-05
PPI	0.08102 ± 0.00003	94.98 [94.34, 95.55]	1.1699 ± 0.0005
PPI++	0.06926 ± 2.3e-06	94.94 [94.30, 95.51]	1.000
PPI++clip	0.06928 ± 2.0e-06	94.92 [94.28, 95.49]	1.0003 ± 1.8e-05
CF1	0.06935 ± 4.4e-06	95.02 [94.38, 95.59]	1.0013 ± 0.0001
CV3	0.06918 ± 3.1e-06	94.92 [94.28, 95.49]	0.9989 ± 3.2e-05
CF3	0.06944 ± 0.00001	95.00 [94.36, 95.57]	1.0027 ± 0.0001
Hoeffding	0.09603 ± 0.00000	99.28 [99.00, 99.48]	1.3866 ± 4.6e-05
EB-label	0.13026 ± 2.6e-06	99.96 [99.85, 99.99]	1.8808 ± 4.3e-05
EB-raw	0.21290 ± 0.00006	100.00 [99.92, 100.00]	3.074 ± 0.001
EB-split1	0.20247 ± 0.00006	100.00 [99.92, 100.00]	2.923 ± 0.001
EB-split3	0.20530 ± 0.00007	100.00 [99.92, 100.00]	2.964 ± 0.001
Exact-binomial	0.07039 ± 1.7e-06	95.70 [95.10, 96.23]	1.0164 ± 2.3e-05

F.7 Two-valued score

Table 20: Two-valued score, $n = 200$, $N = 2000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.13859 ± 0.00001	94.10 [93.41, 94.72]	1.553 ± 0.003
PPI	0.09768 ± 0.00023	94.72 [94.07, 95.31]	1.0946 ± 0.0003
PPI++	0.08924 ± 0.00018	94.50 [93.83, 95.10]	1.000
PPI++clip	0.08924 ± 0.00018	94.50 [93.83, 95.10]	1.000
CF1	0.08963 ± 0.00018	94.54 [93.88, 95.14]	1.0044 ± 0.0002
CV3	0.08924 ± 0.00018	94.50 [93.83, 95.10]	1.000
CF3	0.08963 ± 0.00018	94.54 [93.88, 95.14]	1.0044 ± 0.0002
Hoeffding	0.19206 ± 0.00000	99.60 [99.38, 99.74]	2.152 ± 0.004
EB-label	0.31210 ± 0.00002	100.00 [99.92, 100.00]	3.497 ± 0.007
EB-raw	0.46286 ± 0.00041	100.00 [99.92, 100.00]	5.187 ± 0.006
EB-split1	0.50359 ± 0.00042	100.00 [99.92, 100.00]	5.643 ± 0.009
EB-split3	0.50359 ± 0.00042	100.00 [99.92, 100.00]	5.643 ± 0.009
Exact-binomial	0.14234 ± 0.00001	95.92 [95.34, 96.43]	1.595 ± 0.003

Table 21: Two-valued score, $n = 800$, $N = 8000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.06930 ± 1.6e-06	95.08 [94.45, 95.65]	1.548 ± 0.002
PPI	0.04897 ± 0.00006	95.06 [94.42, 95.63]	1.0938 ± 0.0002
PPI++	0.04477 ± 0.00005	94.94 [94.30, 95.51]	1.000
PPI++clip	0.04477 ± 0.00005	94.94 [94.30, 95.51]	1.000
CF1	0.04482 ± 0.00005	95.02 [94.38, 95.59]	1.0011 ± 4.8e-05
CV3	0.04477 ± 0.00005	94.94 [94.30, 95.51]	1.000
CF3	0.04482 ± 0.00005	95.02 [94.38, 95.59]	1.0011 ± 4.8e-05
Hoeffding	0.09603 ± 0.00000	99.48 [99.24, 99.64]	2.145 ± 0.002
EB-label	0.13026 ± 2.5e-06	99.98 [99.89, 100.00]	2.909 ± 0.003
EB-raw	0.16903 ± 0.00011	100.00 [99.92, 100.00]	3.776 ± 0.002
EB-split1	0.17654 ± 0.00011	100.00 [99.92, 100.00]	3.943 ± 0.003
EB-split3	0.17654 ± 0.00011	100.00 [99.92, 100.00]	3.943 ± 0.003
Exact-binomial	0.07039 ± 1.6e-06	95.76 [95.17, 96.28]	1.572 ± 0.002

F.8 Rare binary

Table 22: Rare binary, $n = 200$, $N = 2000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.01516 ± 0.00034	63.52 [62.18, 64.84]	1.0023 ± 0.0001
PPI	0.08607 ± 0.00010	95.08 [94.45, 95.65]	5.689 ± 0.125
PPI++	0.01513 ± 0.00034	63.48 [62.14, 64.80]	1.000
PPI++clip	0.01515 ± 0.00034	63.52 [62.18, 64.84]	1.0012 ± 0.0001
CF1	0.01520 ± 0.00034	63.50 [62.16, 64.82]	1.0050 ± 0.0002
CV3	0.01506 ± 0.00034	63.46 [62.12, 64.78]	0.9955 ± 0.0002
CF3	0.01529 ± 0.00034	63.46 [62.12, 64.78]	1.0109 ± 0.0004
Hoeffding	0.10103 ± 0.00014	100.00 [99.92, 100.00]	6.678 ± 0.142
EB-label	0.06783 ± 0.00039	100.00 [99.92, 100.00]	4.484 ± 0.075
EB-raw	0.21687 ± 0.00062	100.00 [99.92, 100.00]	14.336 ± 0.314
EB-split1	0.16750 ± 0.00048	100.00 [99.92, 100.00]	11.072 ± 0.221
EB-split3	0.16821 ± 0.00049	100.00 [99.92, 100.00]	11.119 ± 0.221
Exact-binomial	0.02648 ± 0.00021	97.94 [97.51, 98.30]	1.750 ± 0.026

Table 23: Rare binary, $n = 800$, $N = 8000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.00940 ± 0.00007	90.88 [90.05, 91.65]	1.0006 ± 2.1e-05
PPI	0.04308 ± 0.00002	94.74 [94.09, 95.33]	4.584 ± 0.035
PPI++	0.00940 ± 0.00007	90.88 [90.05, 91.65]	1.000
PPI++clip	0.00940 ± 0.00007	90.88 [90.05, 91.65]	1.0003 ± 1.7e-05
CF1	0.00941 ± 0.00007	90.84 [90.01, 91.61]	1.0013 ± 0.0001
CV3	0.00939 ± 0.00007	90.78 [89.95, 91.55]	0.9988 ± 3.4e-05
CF3	0.00942 ± 0.00007	90.80 [89.97, 91.57]	1.0027 ± 0.0001
Hoeffding	0.05302 ± 0.00007	100.00 [99.92, 100.00]	5.641 ± 0.038
EB-label	0.02490 ± 0.00012	100.00 [99.92, 100.00]	2.650 ± 0.008
EB-raw	0.08004 ± 0.00031	100.00 [99.92, 100.00]	8.516 ± 0.066
EB-split1	0.05058 ± 0.00015	100.00 [99.92, 100.00]	5.382 ± 0.029
EB-split3	0.05073 ± 0.00015	100.00 [99.92, 100.00]	5.397 ± 0.029
Exact-binomial	0.01110 ± 0.00007	95.94 [95.36, 96.45]	1.181 ± 0.002

F.9 Galaxies

Table 24: Galaxies, $n = 200$, $N = 2000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.12136 ± 0.00013	94.74 [94.09, 95.33]	1.411 ± 0.002
PPI	0.08646 ± 0.00017	95.02 [94.38, 95.59]	1.0055 ± 0.0003
PPI++	0.08598 ± 0.00016	94.46 [93.79, 95.06]	1.000
PPI++clip	0.08590 ± 0.00016	94.48 [93.81, 95.08]	0.9991 ± 0.0001
CF1	0.08598 ± 0.00016	95.00 [94.36, 95.57]	0.9999 ± 0.0002
CV3	0.08592 ± 0.00016	93.54 [92.82, 94.19]	0.9993 ± 0.0003
CF3	0.08669 ± 0.00017	94.90 [94.25, 95.48]	1.0082 ± 0.0003
Hoeffding	0.19206 ± 0.00000	99.84 [99.68, 99.92]	2.234 ± 0.004
EB-label	0.28607 ± 0.00020	100.00 [99.92, 100.00]	3.327 ± 0.006
EB-raw	0.42936 ± 0.00030	100.00 [99.92, 100.00]	4.994 ± 0.006
EB-split1	0.48561 ± 0.00044	100.00 [99.92, 100.00]	5.648 ± 0.008
EB-split3	0.49100 ± 0.00050	100.00 [99.92, 100.00]	5.710 ± 0.008
Exact-binomial	0.12538 ± 0.00013	95.78 [95.19, 96.30]	1.458 ± 0.002

Table 25: Galaxies, $n = 800$, $N = 8000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.06073 ± 0.00003	94.46 [93.79, 95.06]	1.412 ± 0.001
PPI	0.04330 ± 0.00004	95.26 [94.64, 95.82]	1.0072 ± 0.0001
PPI++	0.04299 ± 0.00004	95.08 [94.45, 95.65]	1.000
PPI++clip	0.04299 ± 0.00004	95.08 [94.45, 95.65]	1.0000 ± 1.2e-05
CF1	0.04299 ± 0.00004	95.42 [94.80, 95.97]	0.9999 ± 0.0001
CV3	0.04299 ± 0.00004	94.70 [94.04, 95.29]	0.9998 ± 0.0001
CF3	0.04307 ± 0.00004	95.14 [94.51, 95.70]	1.0019 ± 0.0001
Hoeffding	0.09603 ± 0.00000	99.84 [99.68, 99.92]	2.234 ± 0.002
EB-label	0.11732 ± 0.00005	99.98 [99.89, 100.00]	2.729 ± 0.002
EB-raw	0.15218 ± 0.00007	100.00 [99.92, 100.00]	3.540 ± 0.002
EB-split1	0.16985 ± 0.00009	100.00 [99.92, 100.00]	3.951 ± 0.002
EB-split3	0.17067 ± 0.00009	100.00 [99.92, 100.00]	3.970 ± 0.002
Exact-binomial	0.06186 ± 0.00003	95.38 [94.76, 95.93]	1.439 ± 0.001

F.10 Forest

Table 26: Forest, $n = 200$, $N = 2000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.09918 ± 0.00019	95.02 [94.38, 95.59]	1.163 ± 0.002
PPI	0.08584 ± 0.00019	94.90 [94.25, 95.48]	1.0067 ± 0.0004
PPI++	0.08527 ± 0.00019	94.38 [93.71, 94.98]	1.000
PPI++clip	0.08510 ± 0.00019	94.42 [93.75, 95.02]	0.9981 ± 0.0002
CF1	0.08543 ± 0.00019	94.66 [94.00, 95.25]	1.0019 ± 0.0003
CV3	0.08512 ± 0.00019	92.92 [92.18, 93.60]	0.998 ± 0.001
CF3	0.08590 ± 0.00019	94.48 [93.81, 95.08]	1.007 ± 0.001
Hoeffding	0.19200 ± 0.00002	100.00 [99.92, 100.00]	2.252 ± 0.005
EB-label	0.25181 ± 0.00034	100.00 [99.92, 100.00]	2.953 ± 0.004
EB-raw	0.36015 ± 0.00073	100.00 [99.92, 100.00]	4.224 ± 0.005
EB-split1	0.39023 ± 0.00085	100.00 [99.92, 100.00]	4.577 ± 0.007
EB-split3	0.39417 ± 0.00087	100.00 [99.92, 100.00]	4.623 ± 0.007
Exact-binomial	0.10357 ± 0.00019	96.44 [95.89, 96.92]	1.215 ± 0.002

Table 27: Forest, $n = 800$, $N = 8000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.04969 ± 0.00005	95.62 [95.02, 96.15]	1.164 ± 0.001
PPI	0.04306 ± 0.00005	94.84 [94.19, 95.42]	1.0086 ± 0.0002
PPI++	0.04269 ± 0.00005	94.66 [94.00, 95.25]	1.000
PPI++clip	0.04269 ± 0.00005	94.66 [94.00, 95.25]	0.9999 ± 2.1e-05
CF1	0.04271 ± 0.00005	94.96 [94.32, 95.53]	1.0004 ± 0.0001
CV3	0.04254 ± 0.00005	94.40 [93.73, 95.00]	0.9965 ± 0.0001
CF3	0.04264 ± 0.00005	94.78 [94.13, 95.36]	0.9987 ± 0.0001
Hoeffding	0.09603 ± 0.00000	99.96 [99.85, 99.99]	2.249 ± 0.002
EB-label	0.10065 ± 0.00007	99.98 [99.89, 100.00]	2.358 ± 0.002
EB-raw	0.14585 ± 0.00008	100.00 [99.92, 100.00]	3.416 ± 0.002
EB-split1	0.16325 ± 0.00010	100.00 [99.92, 100.00]	3.824 ± 0.003
EB-split3	0.16373 ± 0.00011	100.00 [99.92, 100.00]	3.835 ± 0.003
Exact-binomial	0.05087 ± 0.00005	96.06 [95.48, 96.56]	1.192 ± 0.001

F.11 Protein disorder

Table 28: Protein disorder, $n = 200$, $N = 2000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.10579 ± 0.00017	94.52 [93.85, 95.12]	1.584 ± 0.004
PPI	0.06640 ± 0.00019	94.70 [94.04, 95.29]	0.9943 ± 0.0005
PPI++	0.06678 ± 0.00018	93.54 [92.82, 94.19]	1.000
PPI++clip	0.06641 ± 0.00018	93.82 [93.12, 94.45]	0.9944 ± 0.0004
CF1	0.06636 ± 0.00018	94.54 [93.88, 95.14]	0.9936 ± 0.0005
CV3	0.06688 ± 0.00019	91.48 [90.67, 92.22]	1.001 ± 0.001
CF3	0.06716 ± 0.00019	94.50 [93.83, 95.10]	1.006 ± 0.001
Hoeffding	0.19206 ± 7.0e-07	99.96 [99.85, 99.99]	2.876 ± 0.008
EB-label	0.26248 ± 0.00027	100.00 [99.92, 100.00]	3.930 ± 0.009
EB-raw	0.37243 ± 0.00054	100.00 [99.92, 100.00]	5.577 ± 0.010
EB-split1	0.40220 ± 0.00066	100.00 [99.92, 100.00]	6.023 ± 0.012
EB-split3	0.40585 ± 0.00070	100.00 [99.92, 100.00]	6.077 ± 0.013
Exact-binomial	0.11006 ± 0.00017	96.24 [95.68, 96.73]	1.648 ± 0.004

Table 29: Protein disorder, $n = 800$, $N = 8000$.

Method	Width	Coverage (%)	Width / PPI++
Classical	0.05291 ± 0.00004	95.24 [94.61, 95.80]	1.588 ± 0.002
PPI	0.03334 ± 0.00005	95.04 [94.40, 95.61]	1.0009 ± 0.0001
PPI++	0.03331 ± 0.00004	94.92 [94.28, 95.49]	1.000
PPI++clip	0.03328 ± 0.00004	95.00 [94.36, 95.57]	0.9992 ± 0.0001
CF1	0.03326 ± 0.00004	95.20 [94.57, 95.76]	0.9984 ± 0.0001
CV3	0.03328 ± 0.00005	94.30 [93.62, 94.91]	0.9991 ± 0.0002
CF3	0.03332 ± 0.00005	94.92 [94.28, 95.49]	1.0003 ± 0.0002
Hoeffding	0.09603 ± 0.00000	99.98 [99.89, 100.00]	2.883 ± 0.004
EB-label	0.10551 ± 0.00007	100.00 [99.92, 100.00]	3.168 ± 0.004
EB-raw	0.13345 ± 0.00008	100.00 [99.92, 100.00]	4.006 ± 0.003
EB-split1	0.15046 ± 0.00010	100.00 [99.92, 100.00]	4.517 ± 0.004
EB-split3	0.15136 ± 0.00011	100.00 [99.92, 100.00]	4.544 ± 0.004
Exact-binomial	0.05407 ± 0.00004	95.38 [94.76, 95.93]	1.623 ± 0.002

G Reproducibility, compute, and research integrity

The source archive contains `code/methods.py` (all intervals), `code/experiment.py` (all repeated-sampling suites and oracle moments), `code/summarize.py` (every table and figure), and `code/verify.py` (scientific implementation checks). A baseline augmentation script regenerated the *same* random draws to add Hoeffding and exact binomial comparators after the main run; it did not alter existing endpoints. A fresh run of the current experiment driver includes them directly. These extra baselines were added to strengthen a finite-sample comparison that would otherwise use unnecessarily conservative binary baselines. The original full-run results and the added baselines share the recorded seeds.

Run the main, ablation, and ratio suites with `-reps 5000 -workers 4`; run the finite suite with `-reps 3000 -workers 4`, then run the summary and validation scripts. The README gives exact shell commands, pinned package versions, data acquisition commands, source checksums, and the table/figure mapping. Seed derivation uses the first eight little-endian bytes of SHA256 applied to a documented fixed prefix and a condition key; it is independent of worker scheduling. The complete repetition-level outputs are compressed CSV files. Results are reproducible up to floating-point linear-algebra differences; no statistical claims depend on an unreported random seed.

The measured environment used Python 3.11.2, a four-core CPU quota on Intel Xeon Gold 6230 hardware, and a 16 GiB memory limit, with one BLAS thread per process. No GPU or paid API was used. Per-suite measured wall times and maximum worker resident memory are below. Some baseline augmentation overlapped the ratio suite, so these times describe the actual run rather than an ideal sequential benchmark. The original main run's per-cell CPU counter was cumulative within a worker; it is not summed as a per-cell CPU measurement. The driver was corrected for subsequent runs. The wall-time fields are unaffected.

Table 30: Measured main scientific runs; setup, PDF compilation, smoke tests, and baseline augmentation are additional small tasks recorded in `logs/`. The maximum RSS is for one worker, not the sum across workers.

Suite	Cells	Repetitions/cell	Wall seconds	Max worker RSS (MiB)
ablation	9	5000	56.7	114.6
finite	6	3000	184.0	125.0
main	22	5000	187.2	121.5
ratio	12	5000	155.8	120.4

A small smoke run checked the original methods and data I/O before the main comparison. It included a fixed finite menu of score helpers with a Bonferroni penalty; we abandoned this route in favor of pilot-fitted controls, which address the same nuisance-regression question more directly. Smoke outputs and the implementation are retained, and were not pooled with final results. There were no numerical failures in the reported suites; adverse coverage is an experimental result, not a discarded run. Initial missing-package errors and unsuccessful guesses of official template URLs are recorded in `logs/RESEARCH_LOG.md`.

To respect the independent-research restriction, we used no topic search engine, citation graph, aggregator, or related-paper service. Literature access was restricted to the supplied starter PDF URLs and the classical empirical-Bernstein source; repository access was pinned to the supplied PPI version. Only official conference/template/ethics pages and data download URLs were additionally accessed. No encounter with the prohibited work was identified, and no effort was made to identify its title, authors, or findings. These access constraints necessarily limit any claim about novelty. The NeurIPS 2026 official style is unchanged. The main text ends before the references and stays within its nine-page limit; references, appendices, and the checklist are excluded under the official instructions.

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Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: Section 6 and Appendix G disclose that an autonomous language-model agent designed the study, wrote code and proofs, interpreted outputs and wrote the report. LLM-generated judgments are not observations in any experiment.

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